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Predict The Warehouse Will Reach Full Capacity

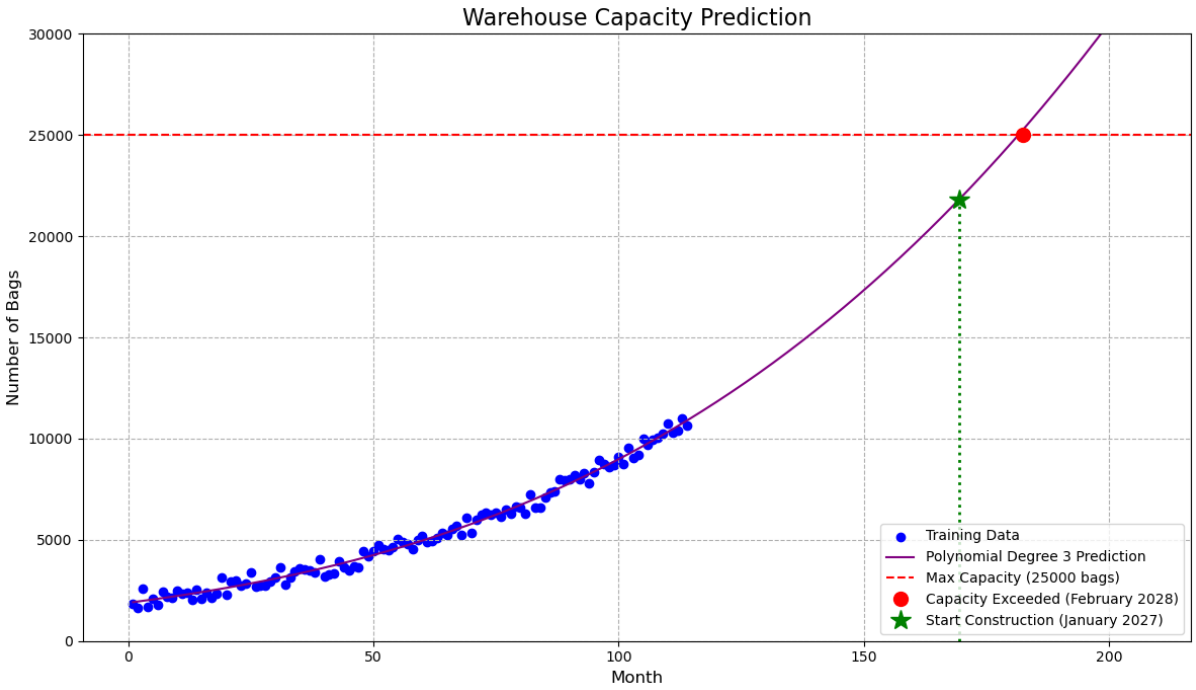
The warehouse was designed to store a maximum of 25,000 (twenty-five thousand) bags each month. We need to predict when the warehouse will reach full capacity.

$$y = 0{,}0016x^3 + 0{,}2298x^2 + 31{,}5678x + 1886{,}7842$$

$$25.000 = 0{,}0016x^3 + 0{,}2298x^2 + 31{,}5678x + 1886{,}7842$$

We cannot manually determine the value of x. By using the Newton-Raphson method, we obtain a result of 182,40. This means that by the 182th month or February 2028, the warehouse will have reached its full capacity of 25,000 bags.

Now we need to predict when to start building a new warehouse. According to the problem, construction must begin 13 months in advance.
x_new_warehouse = 182,40 – 13 = 169,40
This means construction of the new warehouse should begin in the 169th month or January 2027.



Rationale behind using the Newton-Raphson method because it has several advantages, such as the following:

1. Extremely Fast in Finding Solutions
   One of the main strengths of the Newton-Raphson method is its speed. When the initial guess is close to the actual root, this method can converge very quickly. This rapid performance is known as quadratic convergence, meaning that the number of correct decimal places in the result roughly doubles with each iteration.
2. High Precision
   Thanks to its fast convergence, the Newton-Raphson method is highly efficient at achieving

very high levels of accuracy. Even very small error tolerances can be reached within just a few iterations.

3. Flexible for a Wide Range of Equations
   Unlike some other methods that are limited to polynomials, Newton-Raphson is very versatile. It can be applied to many types of non-linear equations, including those involving logarithmic, exponential, and trigonometric functions. This makes it a powerful tool in numerical analysis.

However, the Newton-Raphson method also has some disadvantages, such as:

1. Dependence on the Initial Guess
   This is its biggest drawback. If your initial guess is far from the actual root, the method can:
   - Completely fail (diverge): The result may head toward infinity.
   - Find the wrong root: If there are multiple roots, it might converge to one you didn't intend to find.
   - Oscillate: The output may bounce back and forth without converging to a solution.
2. Requires the Derivative of the Function
   To use the Newton-Raphson method, you must be able to compute the derivative of your function (f'(x)). For some very complex functions, finding the derivative may be difficult or impractical.
   Alternative methods like the Secant Method (similar to Newton-Raphson but without the derivative) or the Bisection Method do not have this requirement.